

Markov Jump Processes

Birth and Death Processes, Asymptotics

ChatGPT (Questions asked by Jan)

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Abstract

These notes give a concise but self-contained account of the material from the slide deck, written in a more formal mathematical style. The emphasis is on a coherent exposition: first the basic construction, then the central examples of birth and death type, and only afterwards the structural results on irreducibility, stationarity, and long-run behavior. Short proof sketches are included where they add clarity without obscuring the main ideas.

1 Setup

Let S be a finite or countable state space. A matrix $Q = (q_{ij})_{i,j \in S}$ is called a Q -matrix if

$$q_{ij} \geq 0 \quad (i \neq j), \quad q_{ii} = - \sum_{j \neq i} q_{ij}.$$

For each $i \in S$ define the total jump rate out of i by

$$\lambda(i) := -q_{ii} = \sum_{j \neq i} q_{ij}.$$

A non-explosive Markov jump process with generator Q is a càdlàg process $(X_t)_{t \geq 0}$ on S whose transition semigroup $(p_t)_{t \geq 0}$ is generated by Q .

For $i \in S$ we write $\mathbb{P}_i$ for the law of the process started from i , and more generally $\mathbb{P}_\mu$ for the law corresponding to an initial distribution μ .

The operational interpretation is as follows. If the process is currently in state i and $\lambda(i) > 0$, then the holding time in i is exponentially distributed with parameter $\lambda(i)$. After this holding time, the process jumps to a new state according to the probabilities

$$P = \{p_{ij}\}_{i,j \in S}, \quad p_{ij} = \frac{q_{ij}}{\lambda(i)} \quad (j \neq i, \lambda(i) > 0).$$

Thus P is the transition matrix of the embedded jump chain $(Y_n)_{n \geq 0}$. If $\lambda(i) = 0$, then state i is absorbing.

There is a one-to-one correspondence between non-explosive Markov jump processes and non-explosive Q -matrices. In practice, the process is therefore described by the pair consisting of the jump probabilities P and the holding-rate function λ .

2 Birth and death processes

Let $S = \mathbb{N}_0$. Given numbers $\beta_i \geq 0$ for $i \in \mathbb{N}_0$ and $\delta_i \geq 0$ for $i \in \mathbb{N}$, define

$$q_{ij} = \begin{cases} \beta_i, & j = i + 1, \\ \delta_i, & i \in \mathbb{N}, j = i - 1, \\ -(\beta_i + \delta_i), & i = j \in \mathbb{N}, \\ -\beta_0, & i = j = 0, \\ 0, & \text{otherwise.} \end{cases}$$

This is the generator of a birth and death process. The number β_i is the birth rate at state i , while δ_i is the death rate at state i for $i \geq 1$.

Under $\mathbb{P}_i$, the local behavior is therefore

$$\begin{aligned} \mathbb{P}_i(X_t = i + 1) &= \beta_i t + o(t), & i \in \mathbb{N}_0, \\ \mathbb{P}_i(X_t = i - 1) &= \delta_i t + o(t), & i \in \mathbb{N}, \end{aligned}$$

and

$$\mathbb{P}_i(X_t = i) = 1 - (\beta_i + \delta_i)t + o(t), \quad i \in \mathbb{N},$$

while at 0 one has

$$\mathbb{P}_0(X_t = 0) = 1 - \beta_0 t + o(t).$$

The holding-time description is especially simple. If $i \geq 1$ and $\beta_i + \delta_i > 0$, then under $\mathbb{P}_i$ the holding time in state i is exponentially distributed with parameter $\beta_i + \delta_i$. After that holding time, the process jumps to $i + 1$ with probability

$$\frac{\beta_i}{\beta_i + \delta_i}$$

and to $i - 1$ with probability

$$\frac{\delta_i}{\beta_i + \delta_i}.$$

At state 0, if $\beta_0 > 0$, the holding time is $\text{Exp}(\beta_0)$ and the next state is necessarily 1. If $\beta_0 = 0$, then 0 is absorbing. More generally, any state $i \geq 1$ with $\beta_i + \delta_i = 0$ is absorbing.

Remark 1. Birth and death processes need not be non-explosive. In general one must impose, or verify, a non-explosion criterion separately. The present description only identifies the generator and the local dynamics.

3 Pure birth processes

A pure birth process is obtained by setting $\delta_i = 0$ for all $i \geq 1$. Then

$$q_{n,n+1} = \beta_n, \quad q_{nn} = -\beta_n, \quad q_{ij} = 0 \text{ otherwise.}$$

We assume throughout this section that the process is non-explosive.

Fix an initial state $i_0 \in \mathbb{N}_0$. Under $\mathbb{P}_{i_0}$, let $H_1, H_2, \dots$ denote the successive holding times. Thus H_1 is the time spent in state i_0 , H_2 is the time spent in state $i_0 + 1$, and more generally H_n is the time spent in state $i_0 + n - 1$.

Proposition 1. *Under $\mathbb{P}_{i_0}$, the holding times $H_1, H_2, \dots$ are independent and satisfy*

$$H_n \sim \text{Exp}(\beta_{i_0+n-1}), \quad n \geq 1.$$

Consequently, the jump times are given by

$$T_n = H_1 + \dots + H_n, \quad n \geq 1.$$

Proof sketch. At the moment of the n th visit, the process is in state $i_0 + n - 1$, so the next holding time is exponential with parameter β_{i_0+n-1} . Since after each jump the process restarts from the new state, repeated use of the strong Markov property yields independence of the successive holding times. $\square$

Although we are assuming non-explosion in this section, it is useful to record the standard criterion.

Theorem 1 (Explosion criterion for pure birth processes). *Assume $\beta_n > 0$ for all n . Then the pure birth process is explosive if and only if*

$$\sum_{n=0}^{\infty} \frac{1}{\beta_n} < \infty.$$

Equivalently, the process is non-explosive if and only if

$$\sum_{n=0}^{\infty} \frac{1}{\beta_n} = \infty.$$

Proof sketch. Under $\mathbb{P}_{i_0}$, the explosion time is

$$\zeta = \sum_{n=1}^{\infty} H_n.$$

By the proposition above, the variables H_n are independent and

$$\mathbb{E}_{i_0}[H_n] = \frac{1}{\beta_{i_0+n-1}}.$$

For sums of independent exponential variables, finiteness of ζ is equivalent to convergence of the corresponding series of means. Since changing finitely many initial terms does not affect convergence, the criterion reduces to the series $\sum_{n \geq 0} 1/\beta_n$. $\square$

4 Poisson process

If $\beta_n \equiv \lambda > 0$, the pure birth process becomes the Poisson process with rate λ . In this case,

$$\sum_{n=0}^{\infty} \frac{1}{\beta_n} = \sum_{n=0}^{\infty} \frac{1}{\lambda} = \infty,$$

so the explosion criterion shows immediately that the process is non-explosive.

Equivalently, under $\mathbb{P}_0$ the holding times $H_1, H_2, \dots$ are independent and identically distributed with law $\text{Exp}(\lambda)$, and therefore only finitely many jumps occur on every bounded time interval.

5 Irreducibility

Definition 1. *The Q -matrix is called irreducible if for every $i, j \in S$ there exist $n \in \mathbb{N}$ and states $i_0, \dots, i_n \in S$ such that $i_0 = i$, $i_n = j$, and*

$$q_{i_{k-1}i_k} > 0 \quad \text{for } k = 1, \dots, n.$$

Thus every state can be reached from every other state through a finite chain of positive-rate transitions. This is the continuous-time analogue of irreducibility for discrete-time Markov chains.

Proposition 2. *If Q is irreducible, then for all $i, j \in S$ and all $t > 0$ one has*

$$p_t(i, j) > 0.$$

Proof sketch. Choose a path $i = i_0, i_1, \dots, i_n = j$ with positive rates along each step. Consider the event that the process makes exactly these first n jumps, that the sum of the first n holding times is at most t , and that the next jump occurs after time t . Every factor entering this event has positive probability, because the jump probabilities along the path are positive and exponential holding times assign positive probability to every interval of positive length. Hence the whole event has positive probability, which implies $p_t(i, j) > 0$. $\square$

In particular, there is no periodicity obstruction in continuous time: randomness in the holding times destroys the rigid lattice phenomenon familiar from discrete-time chains.

6 Stationary distributions

Definition 2. *A probability vector $\pi = (\pi_i)_{i \in S}$ is called a stationary distribution if*

$$\pi p_t = \pi \quad \text{for every } t \geq 0.$$

Theorem 2. *Assume that S is finite and that Q is irreducible. Then the following hold.*

- (1) *There exists a unique stationary distribution.*
- (2) *A probability vector π is a stationary distribution if and only if*

$$\pi Q = 0.$$

Proof sketch. For (1), fix $t > 0$. By irreducibility of Q we know that

$$p_t(i, j) > 0 \quad \text{for all } i, j \in S.$$

Hence the stochastic matrix p_t is irreducible, indeed all its entries are positive. By Theorem 6.22 in the lecture notes, p_t therefore has a unique stationary distribution, say $\pi^{(t)}$. This gives uniqueness of a stationary distribution for the continuous-time chain. For existence, it remains to observe that $\pi^{(t)}$ does not depend on $t > 0$; we omit the proof of this point here.

For (2), suppose first that π is a stationary distribution. Then

$$\pi p_t = \pi p_0 = \pi \quad \text{for all } t \geq 0,$$

so

$$\pi \frac{p_t - p_0}{t} = 0 \quad \text{for all } t > 0.$$

Letting $t \rightarrow 0$ yields $\pi Q = 0$. Since S is finite, there are no convergence issues in passing to the limit.

Conversely, assume that $\pi Q = 0$. Since S is finite, we may write

$$p_t = e^{Qt} = \sum_{n=0}^{\infty} \frac{t^n Q^n}{n!}.$$

Because $\pi Q = 0$, we also have $\pi Q^n = 0$ for every $n \geq 1$. Therefore

$$\pi p_t = \pi \sum_{n=0}^{\infty} \frac{t^n Q^n}{n!} = \pi,$$

so π is stationary. $\square$

Componentwise, the equation $\pi Q = 0$ reads

$$\sum_{i \in S} \pi_i q_{ij} = 0, \quad j \in S,$$

which is the global balance relation.

Remark 2. In practice one usually does not know the matrices p_t explicitly. For that reason, the characterization $\pi Q = 0$ is typically much more useful than the identity $\pi p_t = \pi$.

7 Convergence to equilibrium

Theorem 3 (Limiting transition probabilities). *Assume Q is non-explosive, irreducible, and has stationary distribution π . Then*

$$p_t(i, j) \rightarrow \pi_j \quad \text{as } t \rightarrow \infty$$

for all $i, j \in S$.

We shall not prove this theorem here. Instead, we give the discrete-time intuition behind the result, since it is closely related to the law of large numbers for Markov chains.

Let $(Y_n)_{n \geq 0}$ be the embedded jump chain, and let $H_1, H_2, \dots$ be the successive holding times. The chain (Y_n) records the states visited at jump epochs, whereas the holding times record how long the continuous-time process stays in those states. If the embedded chain spends, in the long run, a certain fraction of its jumps in state j , then the continuous-time process does not necessarily spend the same fraction of physical time in j . States with larger mean holding times are overrepresented when we look at the process at a typical large time.

In the finite-state case this can be seen heuristically from the LLN for Markov chains. If ρ denotes the stationary distribution of the embedded jump chain, then the number of visits to j among the first n jumps is approximately $n\rho_j$. Each such visit has mean length $1/\lambda(j)$. Thus the amount of physical time spent in j is approximately

$$n\rho_j \frac{1}{\lambda(j)},$$

whereas the total elapsed time is approximately

$$\sum_{k \in S} n\rho_k \frac{1}{\lambda(k)}.$$

Consequently, the long-run fraction of physical time spent in j should be

$$\frac{\rho_j/\lambda(j)}{\sum_{k \in S} \rho_k/\lambda(k)}.$$

This is precisely the type of distribution that appears as the stationary distribution π for the continuous-time process. The limiting statement $p_t(i, j) \rightarrow \pi_j$ says that, when the process is observed at a large deterministic time, its distribution is governed by these long-run physical-time proportions. In continuous time there is no additional aperiodicity condition: the exponential holding times smooth out the periodicity phenomena familiar from discrete-time Markov chains.

8 Occupation times

Theorem 4 (Law of large numbers for occupation times). *Assume Q is irreducible and has stationary distribution π . Then for every initial distribution μ and every $j \in S$,*

$$\frac{1}{T} \int_0^T \mathbf{1}_{\{X_t=j\}} dt \rightarrow \pi_j \quad \text{a.s. under } \mathbb{P}_\mu.$$

The statement says that π_j is the long-run fraction of physical time spent in state j . This should again be compared with the discrete-time LLN for Markov chains: in discrete time one counts the proportion of steps with $Y_n = j$, while in continuous time one weights each visit by the corresponding holding time. Thus the continuous-time average is a time-weighted version of the familiar discrete-time occupation average.

9 Visit frequencies

Assume $\lambda(j) > 0$ and let $N_j(T)$ denote the number of visits to state j initiated before time T . Each visit to j contributes a holding time with mean $1/\lambda(j)$. Therefore one expects the asymptotic relation

$$\frac{1}{T} \int_0^T \mathbf{1}_{\{X_t=j\}} dt \approx \frac{N_j(T)}{T} \cdot \frac{1}{\lambda(j)}.$$

Combining this heuristic with the occupation-time theorem suggests that

$$\frac{N_j(T)}{T} \rightarrow \pi_j \lambda(j).$$

This identity can be justified rigorously by renewal arguments. It says that the long-run rate of visits to j equals the stationary mass of j multiplied by the rate at which the process leaves j on each visit.